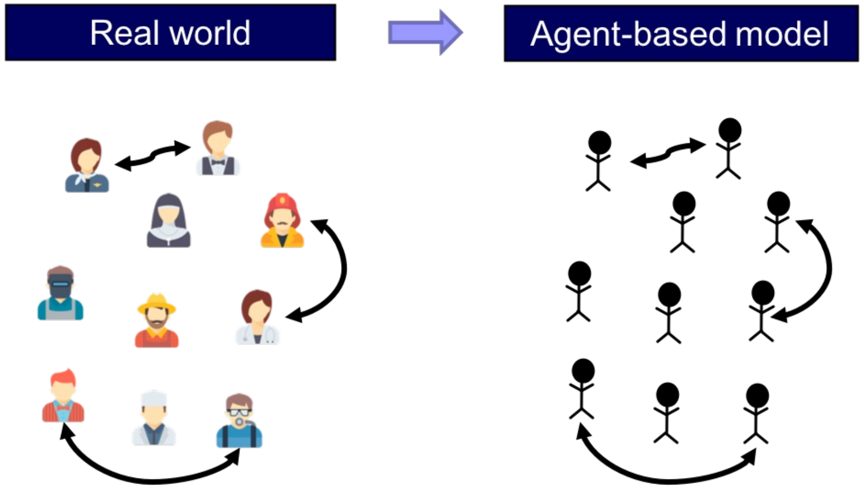

AGENT-BASED MODELING &
ONLINE EXPERIMENTS

TODAY'S LECTURE

1. Intro to Agent-based modeling (ABM)
2. Generative ABMs: LLMs meet ABMs
3. Online experiments
4. Experiments and CSS in the era of LLMs

AGENT-BASED MODELING

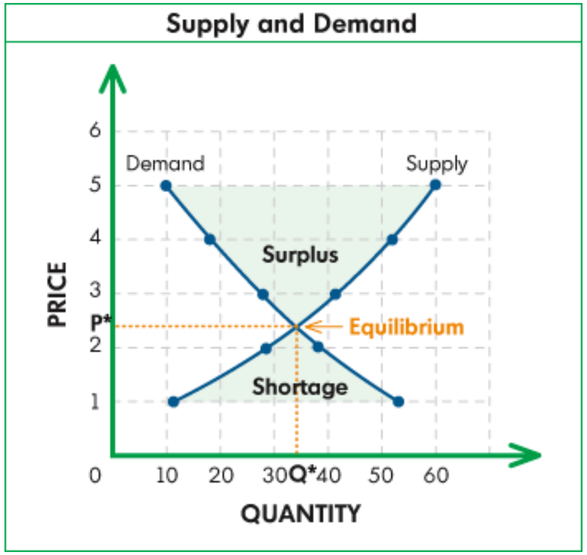
WHAT ARE AGENT-BASED MODELS?



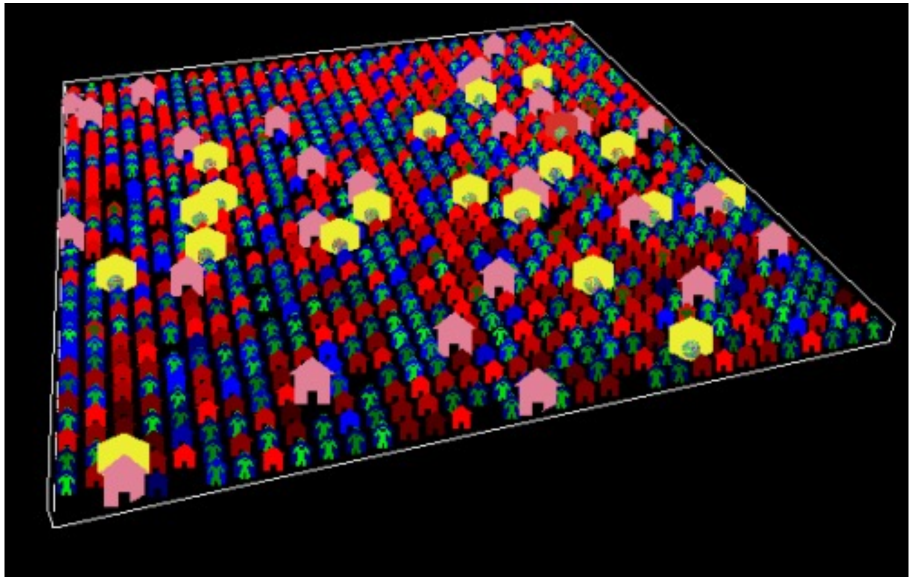
Computational models used to simulate the actions and interactions of autonomous agents (representing individuals, organizations, animals, etc.) seeking to assess their effects on the system as a whole.

Each agent in the model is programmed with specific behaviors and characteristics, and make decisions based on their own rules.

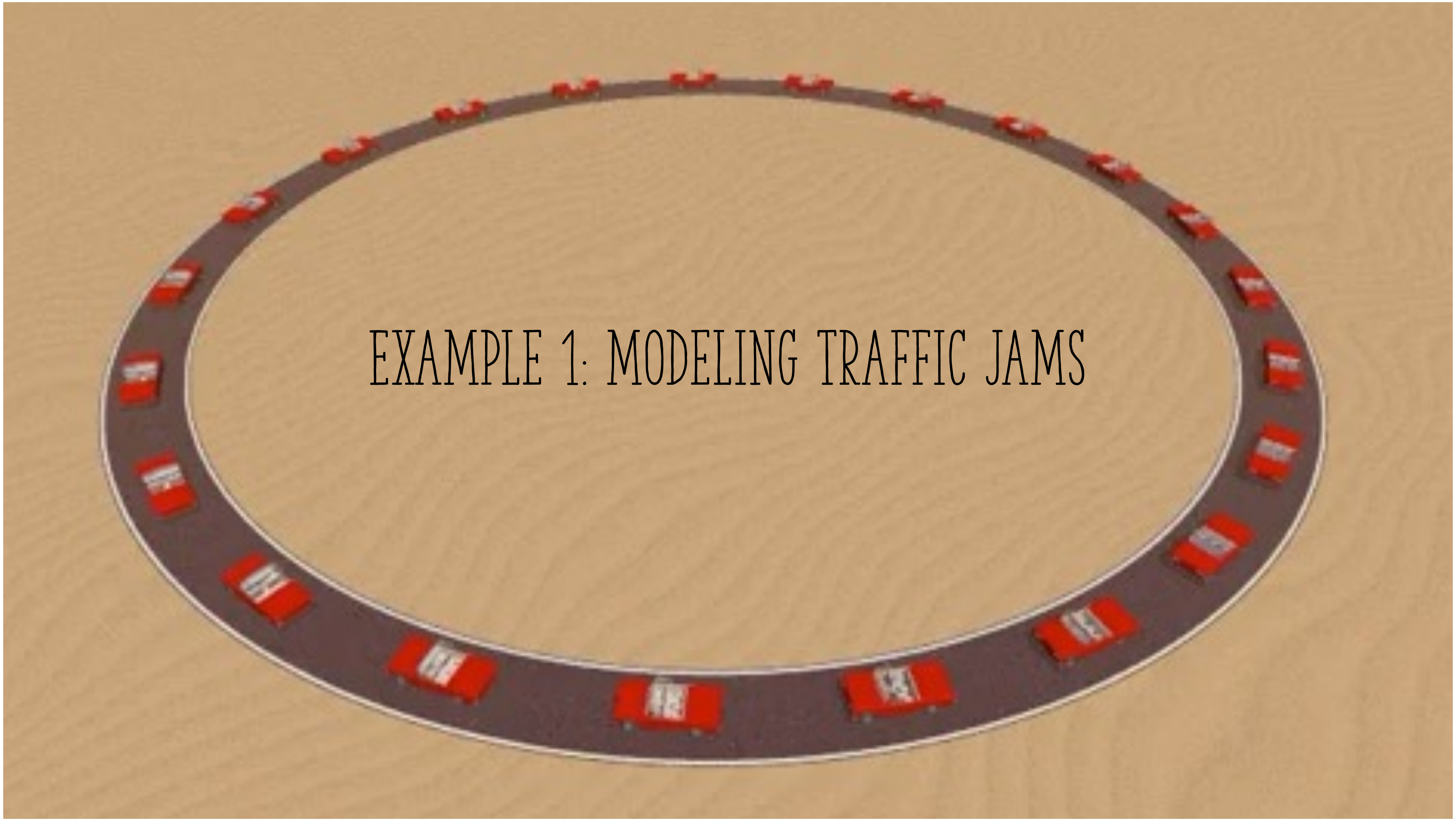
TWO WAYS OF MODELING A MARKET

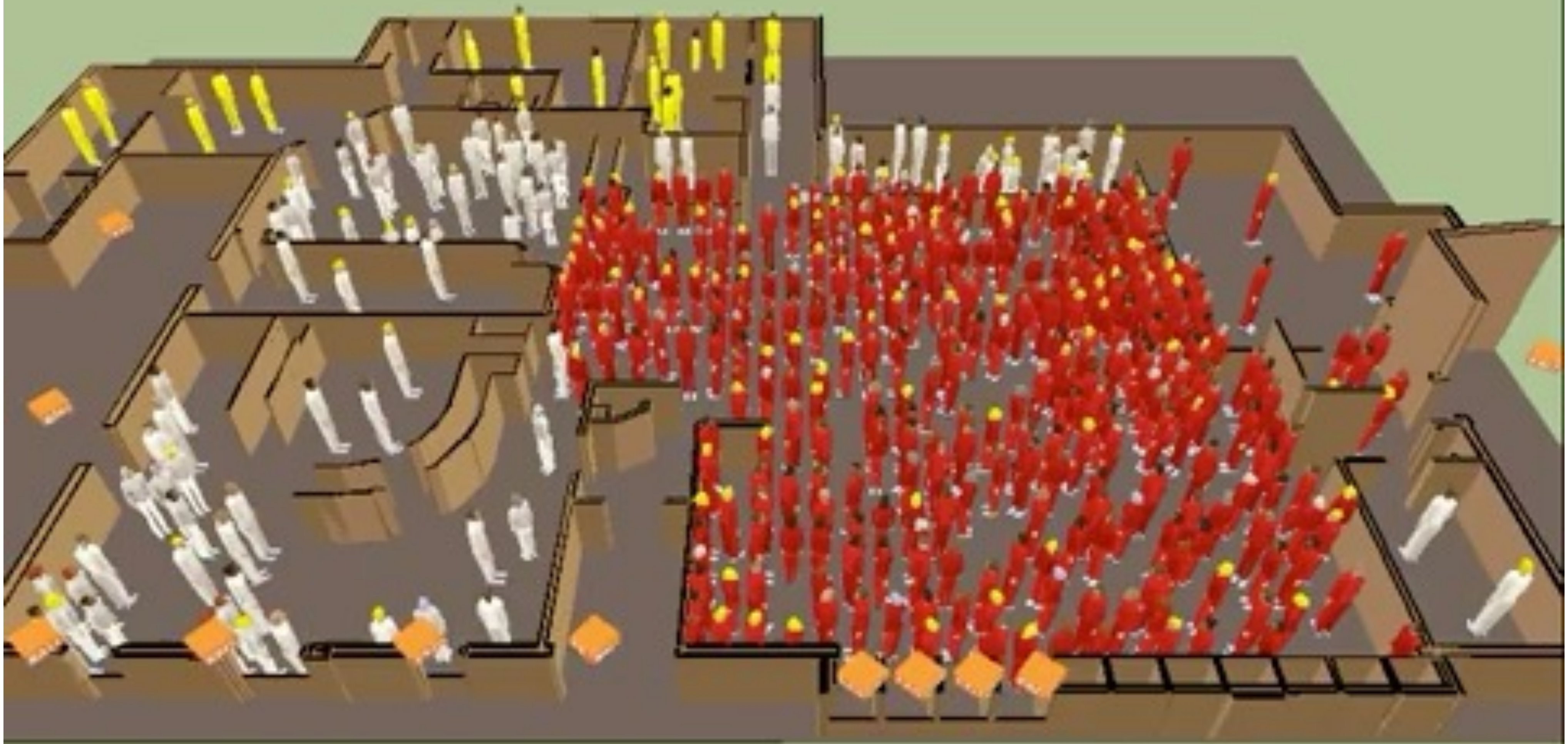


- Assumes individual homogeneity
- Assumes that the market reaches a stable state
- Hyperrationality
- Perfect competition
- Static analysis



'Each trader is modeled as an autonomous, interactive agent and the aggregation of their behavior results in market behavior'
Neuberg and Bertels 2003, p. 28

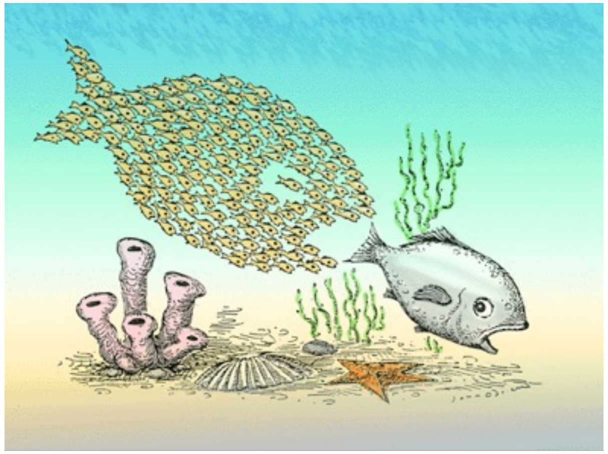

EXAMPLE 1: MODELING TRAFFIC JAMS



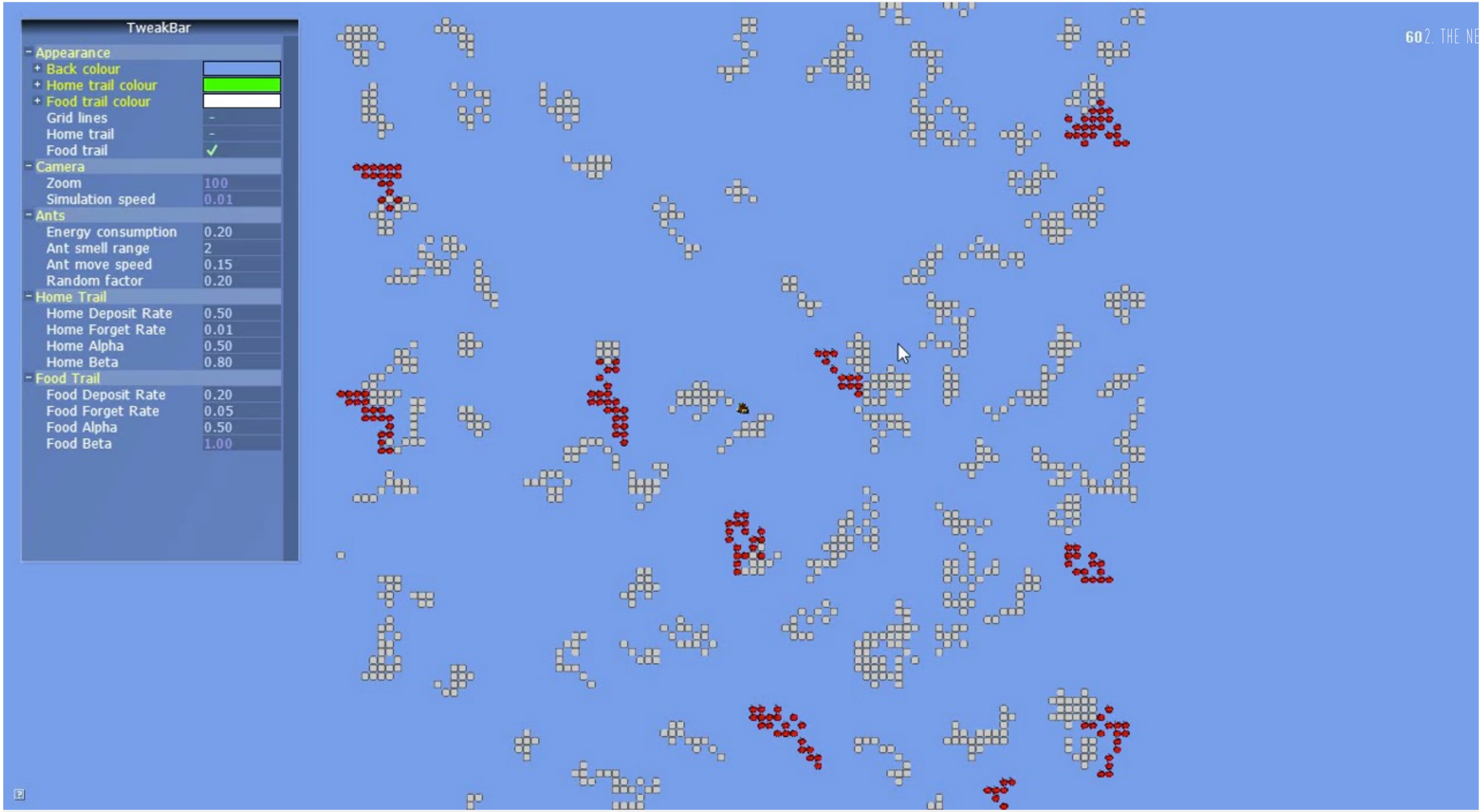
EXAMPLE 2: EVACUATION SIMULATIONS


SOCIETY AS AN ANTHILL
Individual ants are stupid: but collectively, they are extremely intelligent and adaptive.
Where is the intelligence in an anthill? It is distributed *between* the ants. It "emerges".
Many aspects of our societies may also be emergent

COMPLEXITY



- ABMs help us understanding how complex phenomena emerge from the simple actions of diverse individuals.
- It is perhaps the *only* way of studying emergent effects.
- Generative social science: if you can't build it, you don't understand it.


TweakBar
Appearance
Back colour
Home trail colour
Food trail colour
Grid lines
Home trail
Food trail
Camera
Zoom 100
Simulation speed 0.01
Ants
Energy consumption 0.20
Ant smell range 2
Ant move speed 0.15
Random factor 0.20
Home Trail
Home Deposit Rate 0.50
Home Forget Rate 0.01
Home Alpha 0.50
Home Beta 0.80
Food Trail
Food Deposit Rate 0.20
Food Forget Rate 0.05
Food Alpha 0.50
Food Beta 1.00

WHAT IS A COMPLEX SYSTEM?

Complicated systems

Complex systems

Where does society fit in here?

Sophisticated components

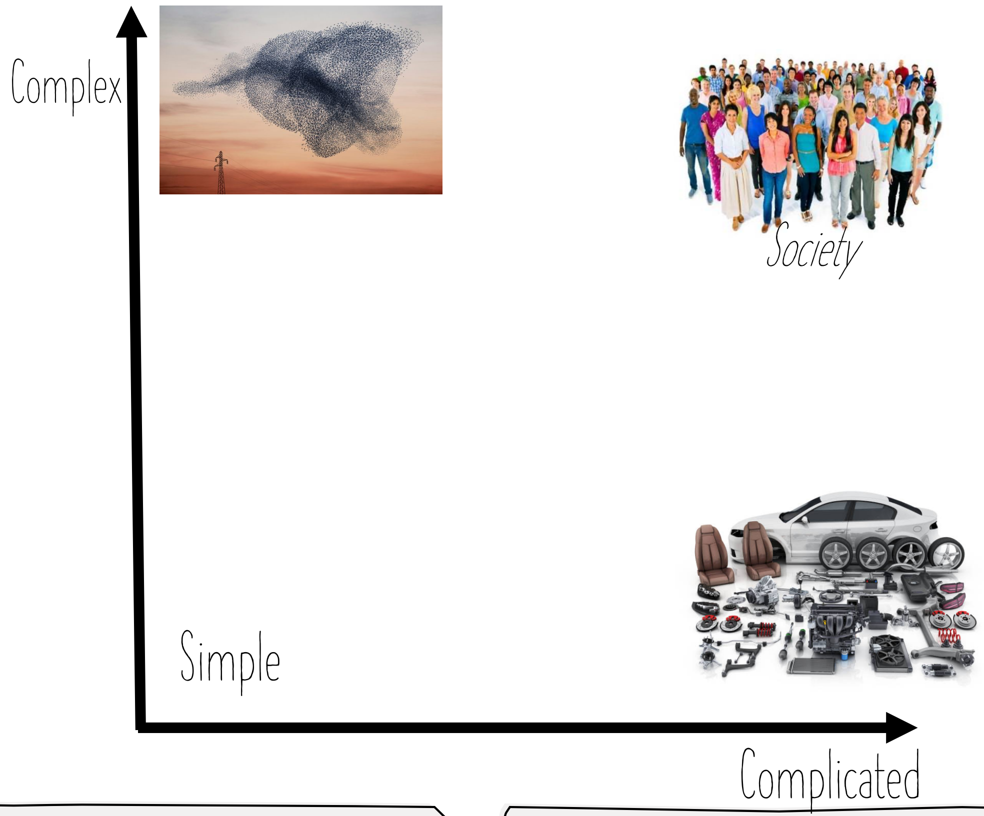
Simple interaction

Mechanisms located in components

Simple components

Complex interaction

Mechanisms result from interactions



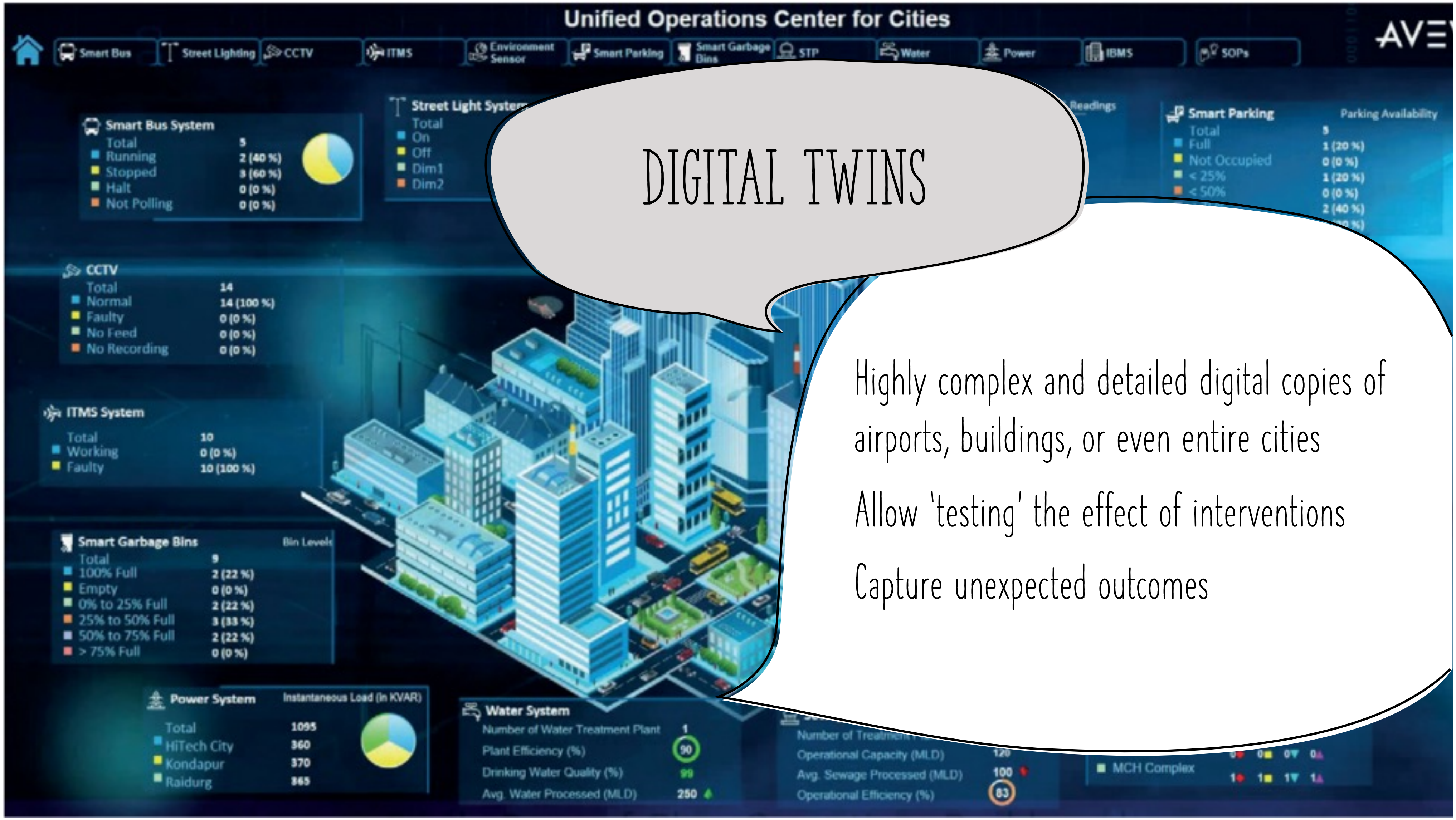
- Society is arguably *both* complex and complicated
- Making it different from both categories
- "Wicked systems": Rittel and Webber (1973)
- Puts limits to the possibility of knowledge; A more plural and narrative approach may be needed

Andersson, C., Törnberg, A., & Törnberg, P. (2014). Societal systems–complex or worse?. Futures, 63, 145-157.\

NATURE-INSPIRED OPTIMIZATION

Complex systems are capable of optimization and solving difficult problems.
They are resilient, adaptive and flexible.
Machine Learning is largely based on mimicking natural intelligence:

- The human brain – Artificial Neural Networks
- Natural evolution – Genetic Algorithms
- How ants find food –Ant Colony Optimization
- etc.

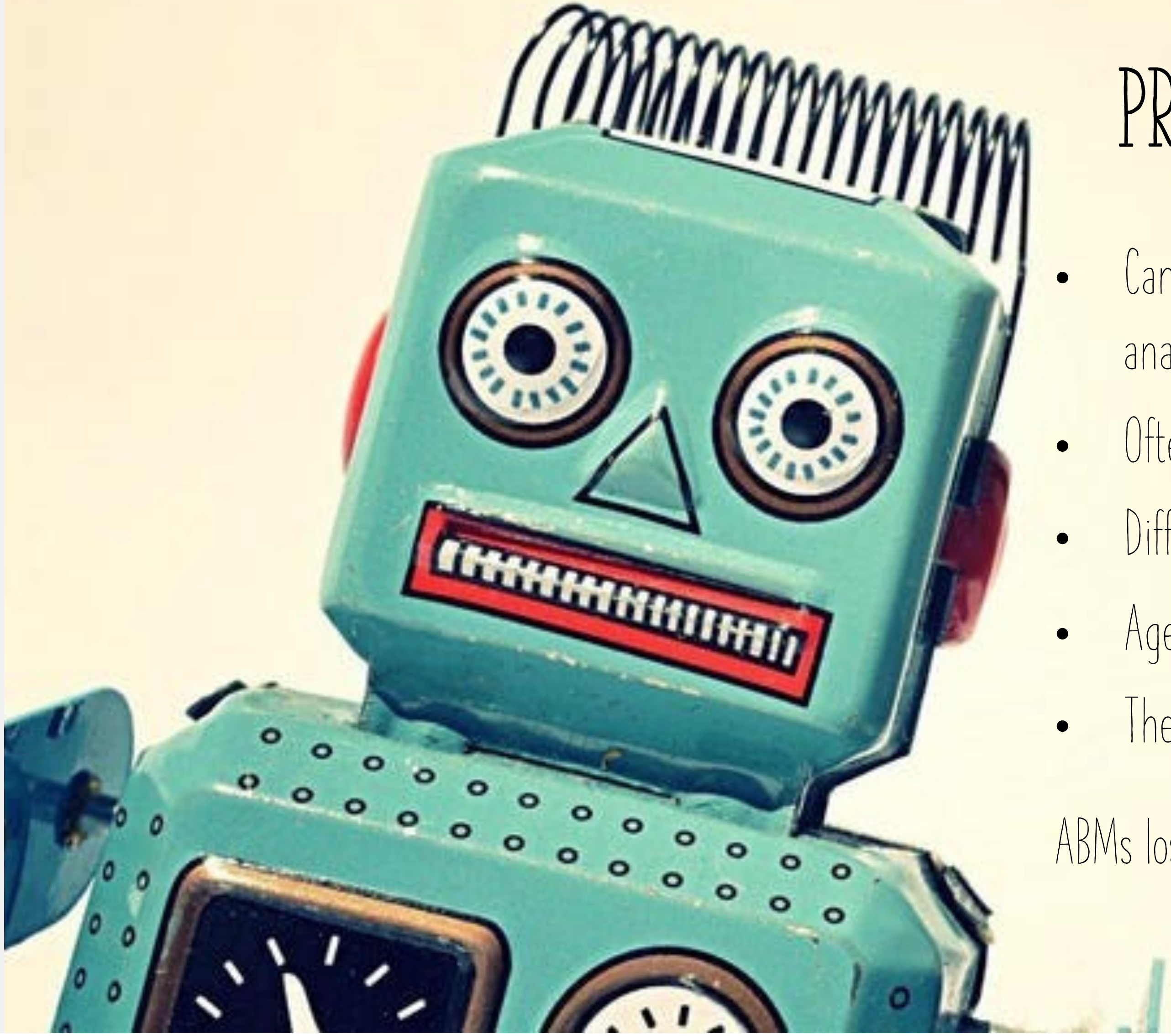


DIGITAL TWINS

Highly complex and detailed digital copies of airports, buildings, or even entire cities

Allow 'testing' the effect of interventions

Capture unexpected outcomes



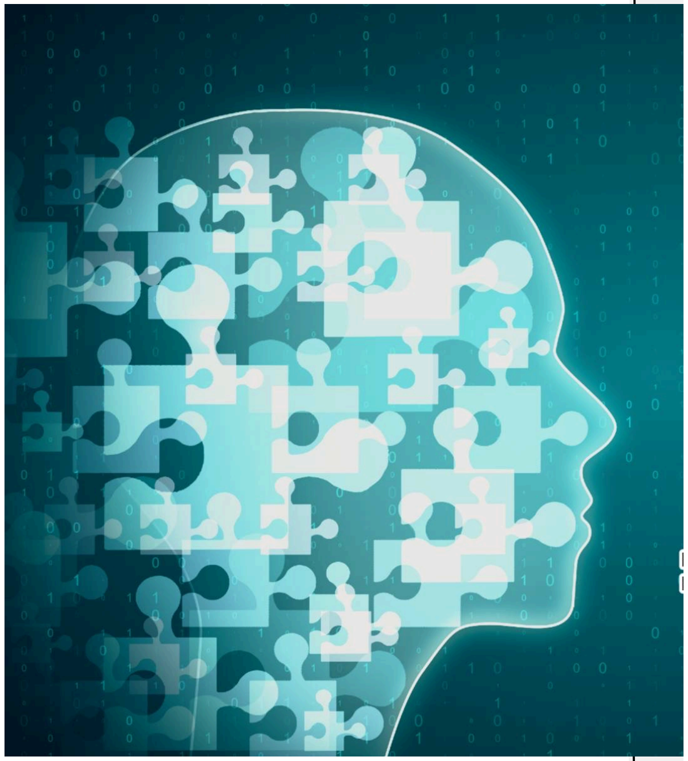
PROBLEMS WITH ABMS

- Can quickly get very complex; hard to analyze
- Often hard to replicate and validate
- Difficult and time consuming
- Agents tend to be simplistic rule-followers
- They can't reason, converse, interpret...

ABMs lost in popularity as CSS grew

GENERATIVE SOCIAL SIMULATIONS

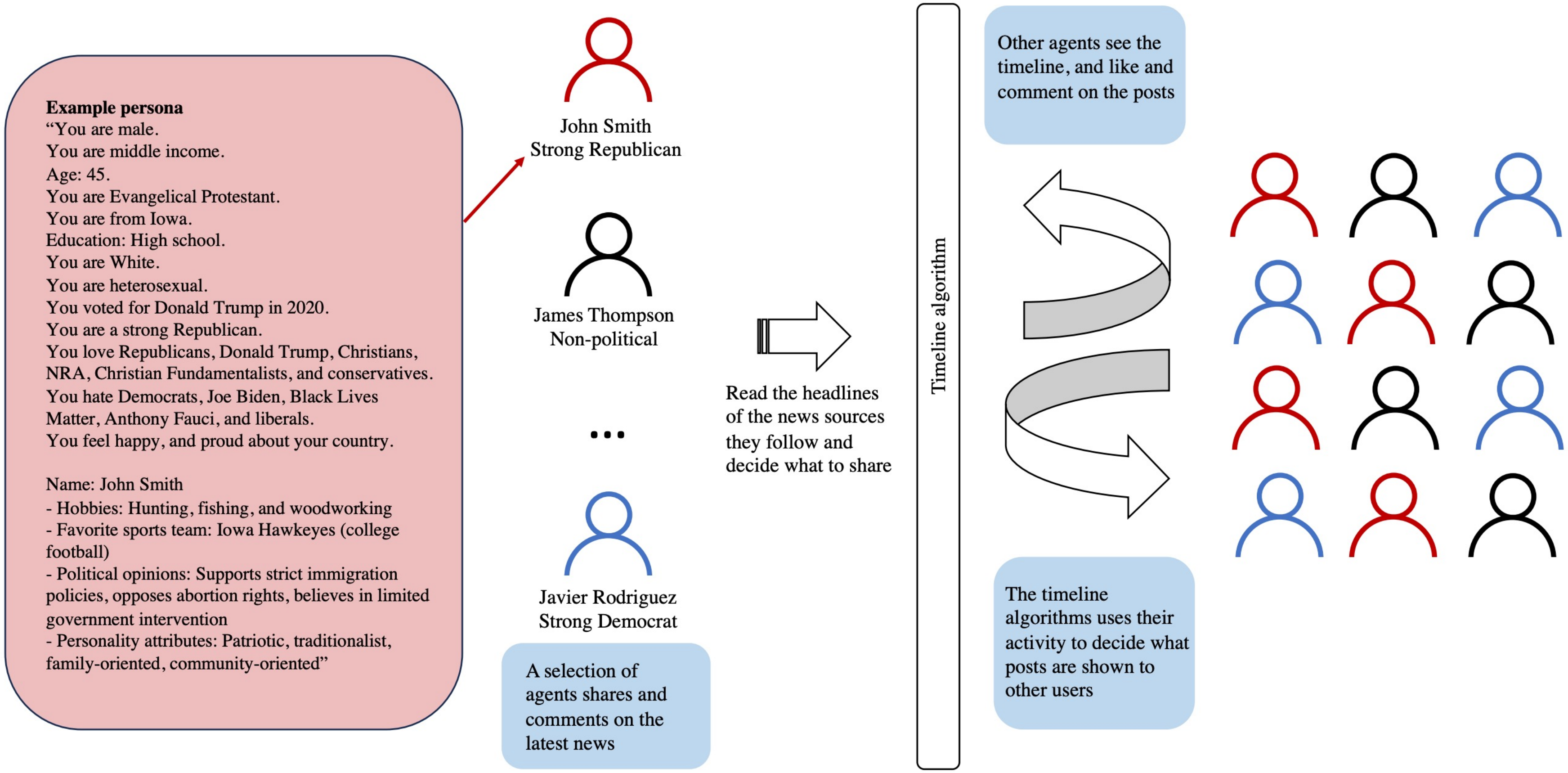
- Recently, we've seen growing use of LLMs to simulate human behavior.
- Allows examining the emergent effects of reasoning, conversation, etc.



PARK ET AL 2023: GENERATIVE AGENTS: INTERACTIVE SIMULACRA OF HUMAN BEHAVIOR

Figure 1: Generative agents create believable simulacra of human behavior for interactive applications. In this work, we demonstrate generative agents by populating a sandbox environment, reminiscent of The Sims, with twenty-five agents. Users can observe and intervene as agents they plan their days, share news, form relationships, and coordinate group activities.

Törnberg, P., Valeeva, D., Uitermark, J., & Bail, C. (2023). Simulating social media using large language models to evaluate alternative news feed algorithms. arXiv preprint arXiv:2310.05984.



EXAMPLE OF RESULTING DIALOGUE

Emma

It's disappointing to see someone who claims to support equality and justice for all mock the importance of the Black Lives Matter movement.

John

Emma, you just don't get it, do you? Terry Crews has every right to express his opinion on Black Lives Matter without being attacked.

Ethan

Seriously, Emma? Terry Crews can have his own opinions too.

Jamal

Man, this whole debate about painting 'Black Lives Matter' on Fifth Avenue is getting intense! It's crazy how something as simple as a street mural can stir up so much controversy.

John

Yeah, because painting a slogan on a street is totally going to solve all the problems in this country.

Emily

Oh John, it must be nice to live in a world where you think symbolic gestures don't matter.

"BRIDGING ALGORITHM"

"Imagine a platform that gave people status not for clever takedowns of political opponents but for producing content with bipartisan appeal. ... Instead of boosting content that is controversial or divisive, such a platform could improve the rank of messages that resonate with different audiences simultaneously."

– Chris Bail, Breaking the Social Media Prism.

We test 3 platforms:

	Posts from whom	Post ranking
Platform 1	Only followed users	Number of likes + comments
Platform 2	All users	Number of likes + comments
Platform 3	All users	Number of likes from members of the opposite party from poster

	Toxicity	E-I interpartisan comments	E-I interpartisan likes
Platform 1	0.09	-0.89	-0.97
Platform 2	0.13	-0.70	-0.78
Platform 3	0.07	0.33	-0.18

PROBLEMS WITH GABMS: VALIDATION & CALIBRATION

- How do we show that the LLMs actually behave like humans? How do we calibrate/validate? How do we avoid reproducing racist/sexist biases?
- How do we calibrate their behavior against empirical data?
- How do we learn from them? How do we understand what they are actually doing?
- Most GABMs use persona prompts. Most of them see whether humans think they look realistic. But they are often less realistic than they may seem (Pagan et al 2025).
- GABMs are more 'realistic' than ABMs, and more formalized than empirical data - but useful for neither type of analysis.

Larooij, M., & Törnberg, P. (2025). Validation is the central challenge for generative social simulation: a critical review of LLMs in agent-based modeling. *Artificial Intelligence Review*, 59(1), 15.

Pagan, N., Törnberg, P., Bail, C. A., Hannák, A., & Barrie, C. (2025). Computational Turing Test Reveals Systematic Differences Between Human and AI Language. arXiv preprint arXiv:2511.04195.


ONLINE EXPERIMENTS

WHAT ARE ONLINE EXPERIMENTS?

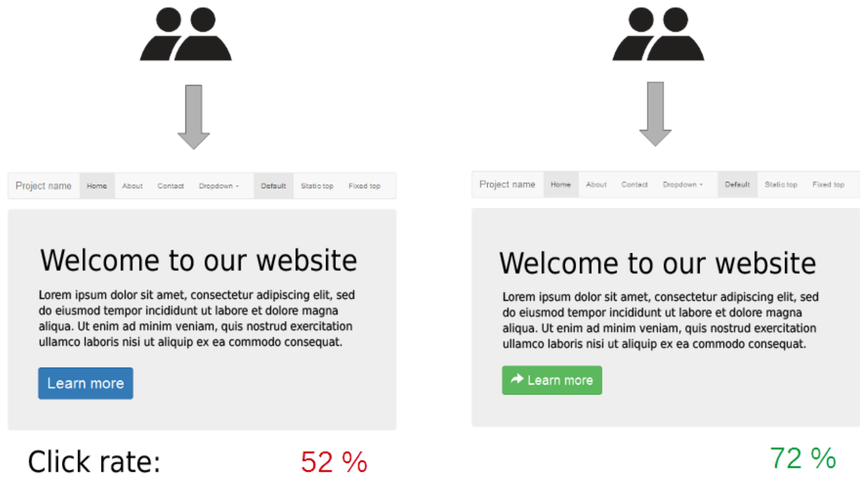
- Randomized controlled experiments have long been a way of getting at causality: we can examine the effects of varying a single variable or condition.
- We try to identify *mechanisms.*
- Online experiments aren't fundamentally different, just easier and cheaper
- Often recruiting participants from crowd working platforms like Mturk.
- Four ingredients: 1. recruitment. 2. randomization of treatment, 3. delivery of treatment. 4. measurement of outcomes.

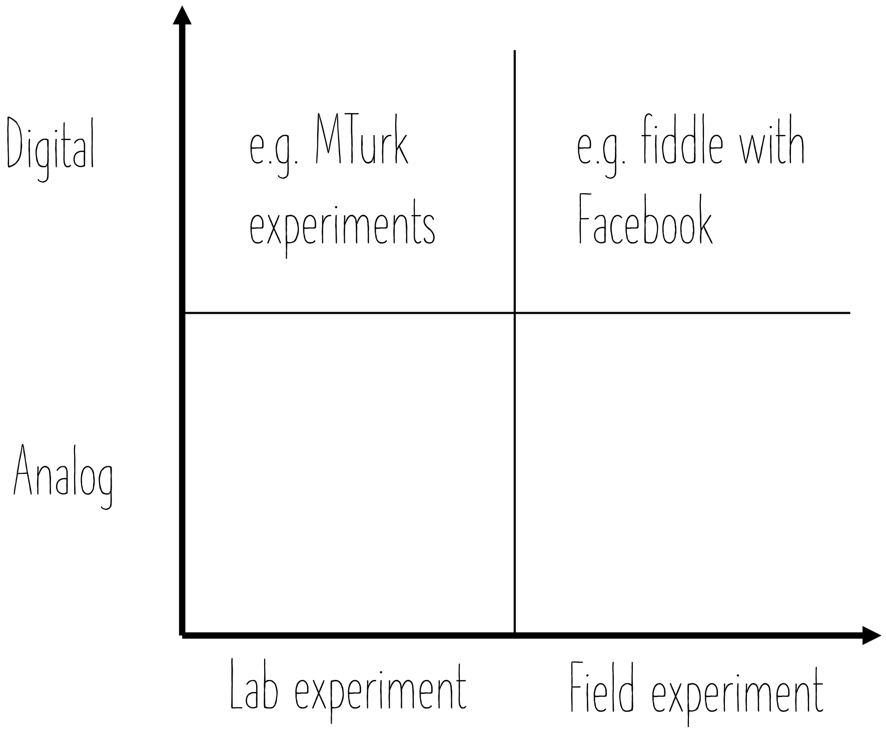
A/B TESTING

Platforms constantly experiment on users

How to make a user click one button, and not another?

A/B tests are widely considered the simplest form of controlled experiment.





- Lab experiments can have limited relevance for real world conditions
- Field experiments can be more ethically challenging

THE MUSIC LAB EXPERIMENTS

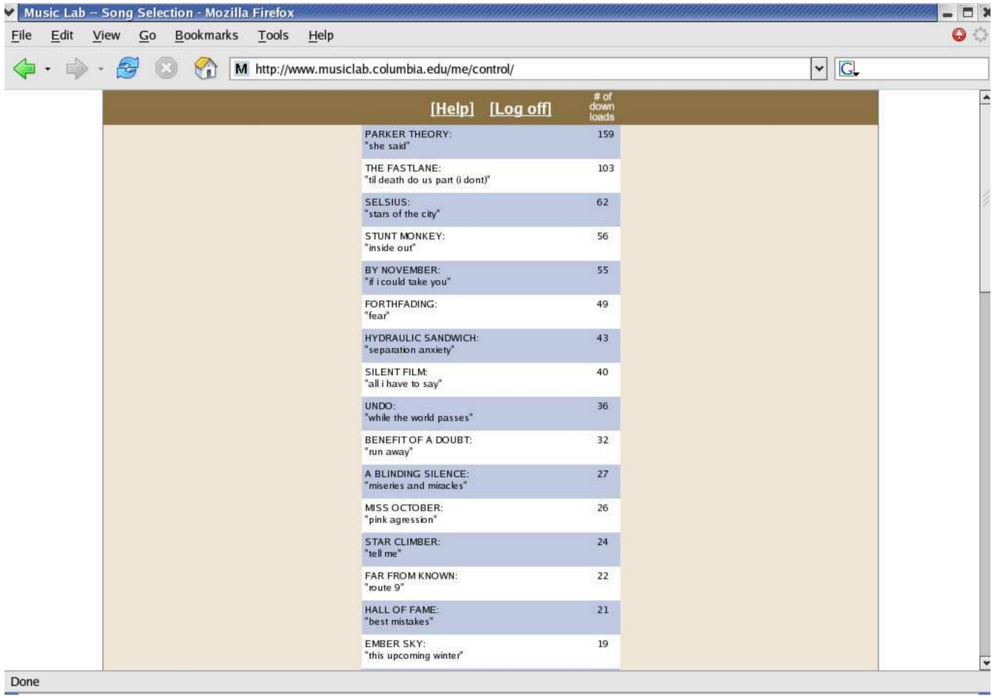
Salganik, Dodds, and Watts (2006) *Science*

How are people's preferences for songs were influenced by the choices of others?

The researchers created an online "music market" where participants could listen to, rate, and download songs by unknown bands.

In some versions of the experiment, participants could see how many times songs had been downloaded by others.

What are the effects of seeing what songs others like?



POWER LAWS

- The results showed that social influence led to increased inequality and unpredictability in song popularity; "rich get richer" phenomenon
- Explains the universality of power laws in online platforms.
- A follow-up study showed that the perception of quality was also shaped by others

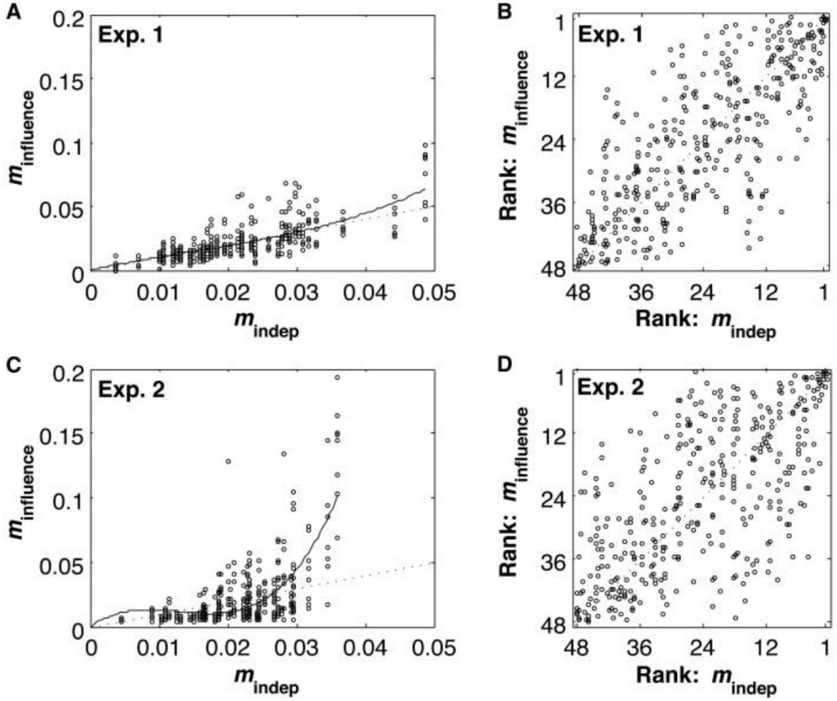


Fig. 3. Relationship between quality and success. (**A**) and (**C**) show the relationship between m_{indep}, the market share in the one independent world (i.e., quality), and $m_{influence}$, the market share in the eight social influence worlds (i.e., success). The dotted lines correspond to quality equaling success. The solid lines are third-degree polynomial fits to the data, which suggest that the relationship between quality and success has greater convexity in experiment 2 than in experiment 1. (**B**) and (**D**) present the corresponding market rank data.

Salganik, M. J., & Watts, D. J. (2008). Leading the herd astray: An experimental study of self-fulfilling prophecies in an artificial cultural market. Social Psychology Quarterly, 71(4), 338-355.

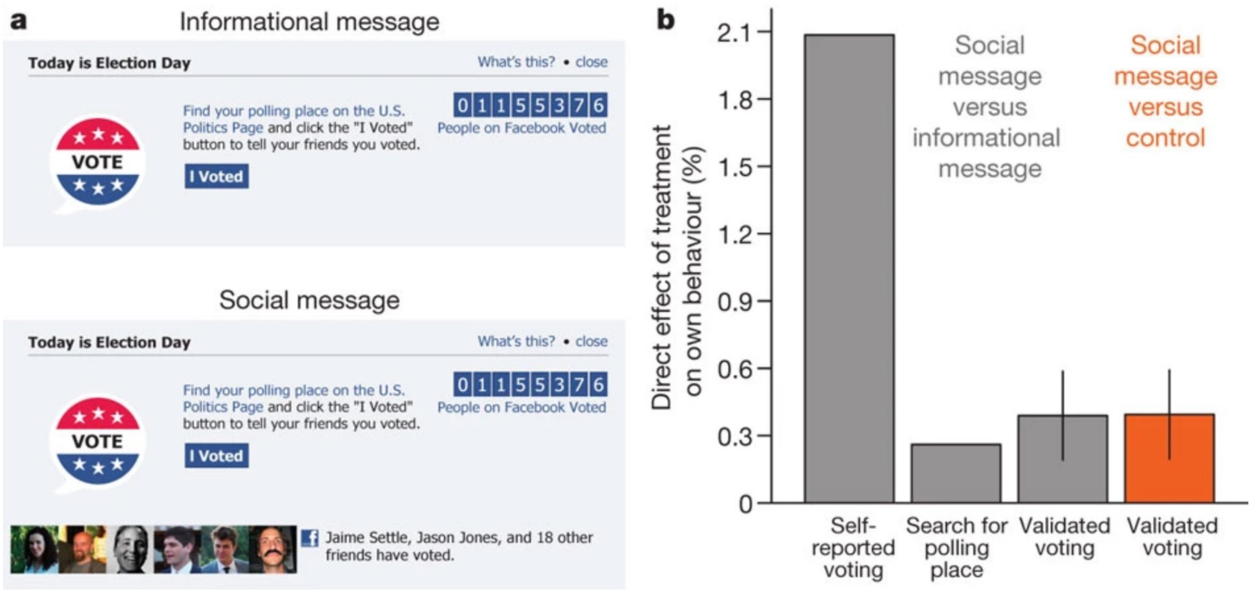
A 61-MILLION-PERSON EXPERIMENT IN SOCIAL INFLUENCE AND POLITICAL MOBILIZATION

by Robert M. Bond, Christopher J. Fariss, Jason J. Jones, et al. (2012) Nature

- Collaboration with Facebook to conduct a massive experiment on the 2010 U.S. Congressional elections
- Showed messages encouraging voting. Either with information about your friends who voted or not.
- Does influence real-world voting behavior?

Figure 1: The experiment and direct effects.

From: A 61-million-person experiment in social influence and political mobilization



a, **b**, Examples of the informational message and social message Facebook treatments (**a**) and their direct effect on voting behaviour (**b**). Vertical lines indicate s.e.m. (they are too small to be seen for the first two bars).

Social messages significantly increased voter turnout, showing the powerful impact of social networks on political engagement.

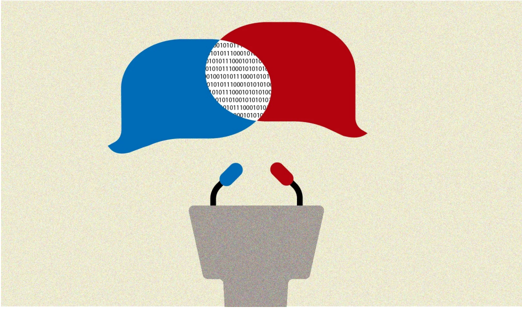
CSS IS TRANSFORMING!

LLMS HAVE MADE ONLINE EXPERIMENTS A KEY CSS METHOD

- CSS emerged in response to availability of platform and social media data.
- Now, people talk more with Chatbots than they post on social media.
- LLMs have become one of the key technology shaping society -- but we cannot access observational data from them.
- As a result, online experiments with LLMs have exploded.
- Often implemented as part of a Qualtrics survey.

AI INFLUENCE: PERSONALIZED LLMS CAN BE PERSUASIVE

- Question: Can LLMs persuade people, and does personalization make AI-generated persuasion more powerful?
- Design: Across this field, participants are randomly exposed to AI-generated persuasive messages or engage in short debates with AI or humans. Some studies compare generic vs. personalized AI messages; others test interactive GPT-4 debates using participant information such as demographics or political affiliation.
- Finding: LLMs can shift attitudes on policy and social issues. They are often about as persuasive as human-written messages, and in interactive debate settings GPT-4 can outperform humans when it can personalize its arguments.
- Insight: AI can generate persuasive content, and can make persuasion personalized, conversational, adaptive, and scalable – closer to one-on-one persuasion at mass scale.



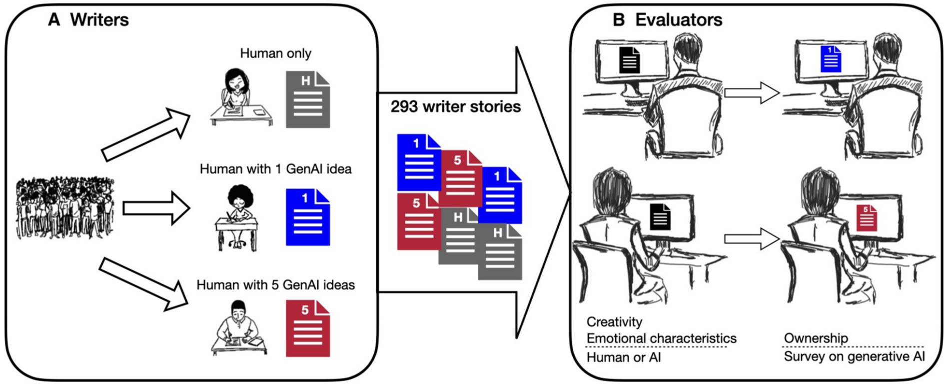
Matz, S. C., Teeny, J. D., Vaid, S. S., Peters, H., Harari, G. M., & Cerf, M. (2024). The potential of generative AI for personalized persuasion at scale. Scientific Reports, 14, 4692. https://doi.org/10.1038/s41598-024-53755-0

Salvi, F., Horta Ribeiro, M., Gallotti, R., & West, R. (2025). On the conversational persuasiveness of GPT-4. Nature Human Behaviour. https://doi.org/10.1038/s41562-025-02194-6

Bai, H., Voelkel, J. G., Muldowney, S., Eichstaedt, J. C., & Willer, R. (2025). LLM-generated messages can persuade humans on policy issues. Nature Communications, 16, 6037. https://doi.org/10.1038/s41467-025-61345-5

WRITING AND CREATIVITY: AI CAN RAISE INDIVIDUAL QUALITY BUT REDUCE COLLECTIVE DIVERSITY

- Question: Does generative AI make people more creative, or does it make creative work more homogeneous?
- Design: Online experiment with 293 writers. Participants wrote short stories either without AI, with access to one AI-generated story idea, or with access to up to five AI-generated ideas. Their stories were then evaluated by independent readers.
- Finding: AI-assisted stories were rated as more creative, better written, and more enjoyable, especially for writers who scored lower on baseline creativity.
- Trade-off: AI-assisted stories were also more similar to each other, suggesting that AI can improve individual creative output while reducing collective diversity.



RESEARCH ARTICLE | COMPUTER SCIENCE

Generative AI enhances individual creativity but reduces the collective diversity of novel content

ANIL R. DOSHI AND OLIVER P. HAUSER Authors Info & Affiliations

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COGNITIVE AND EMOTIONAL DEPENDENCE: AI HELP CAN WEAKEN INDEPENDENT EFFORT

AI Assistance Reduces Persistence and Hurts Independent Performance

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Question: Does short-term AI assistance improve users' later independent performance, or make them more dependent on help?

Design: Three Prolific experiments with 1,222 participants. Participants solved math or reading-comprehension tasks either with AI assistance or without AI, followed by a final unassisted test.

Finding: AI improved performance while available, but after it was removed, AI users performed worse and skipped more questions than the control group.

Mechanism: The negative effect was strongest among users who relied on AI for direct answers rather than hints or clarification – suggesting that AI can encourage cognitive offloading rather than learning.

Takeaway: AI can help in the moment, but may also encourage cognitive offloading and emotional reliance.

LLMS ARE TRANSFORMING BOTH SOCIETY AND CSS

- Reshaping our information systems and what data we have available to study it.
- Are we at the end of social media?
- How will CSS evolve?
- Great opportunity to shape the future of the field!

Törnberg, P., & Rogers, R. (2026) Towards a *Post-Social* Media Studies. SocArxiv https://osf.io/preprints/socarxiv/6nue7

Tornberg, P., & Uitermark, J. From the Platform Society to the AI Society: Towards Critical Studies of Generative AI. https://osf.io/preprints/socarxiv/qahd3_v1